

- RAG Learning

4.1 Energy

AQA GCSE Physics (8463) — Paper 1

4.1.1 Energy Changes in a System

Energy is a property of an object or system. It is *not* a substance — it is a quantity (measured in **joules, J**) that allows us to predict what changes can happen in a system. Energy can be **stored** in various ways and **transferred** between stores. The total amount of energy in a closed system never changes — this is the **principle of conservation of energy**.

AQA GCSE Physics uses the modern "**stores and pathways**" model. Rather than saying energy is "converted" from one type to another, we say energy is **transferred** from one store to another via a pathway. This language may feel unusual, but it is the language examiners expect — answers using it score more marks than older "kinetic energy becomes heat energy" phrasings.

The Eight Energy Stores

All the energy in the universe is held in one of these eight **stores**. You must be able to identify the store(s) involved in any situation.

Store	What is it?	Example
Kinetic	Energy stored in a moving object. Depends on mass and speed.	A car driving, a falling book, a runner, a bullet in flight.
Chemical	Energy stored in the bonds between atoms and molecules. Released in chemical reactions.	Food, fuels (petrol, coal, wood), batteries, explosives.

Gravitational Potential	Energy stored in an object raised up in a gravitational field.	A book on a high shelf, water in a reservoir, a skier at the top of a slope.
Elastic Potential	Energy stored in a stretched or compressed elastic object.	A stretched spring, a drawn bow, a squashed rubber ball, a wound clockwork toy.
Thermal (Internal)	Energy stored in the random movement of the particles making up an object. Hotter object = greater thermal store.	A hot drink, a warm radiator, the inside of an oven.
Nuclear	Energy stored in the nucleus of an atom. Released in fission, fusion or radioactive decay.	Uranium fuel rods in a reactor, the Sun's core, radioactive isotopes.
Electrostatic	Energy stored between two electric charges that attract or repel.	Charged thunder clouds, a charged balloon sticking to a wall, two charged spheres.
Magnetic	Energy stored between two magnetic poles that attract or repel.	Two bar magnets held apart, a fridge magnet pulled off the door.

Exam Tip: Examiners often mark down answers that use old-fashioned terms such as "heat energy", "sound energy", "light energy", "movement energy" or "electrical energy". These are not energy stores — they are **pathways** (ways energy is transferred). Always use one of the eight store names above when describing *where* energy is held.

The Four Energy Pathways (Transfers)

Energy moves between stores via one of these four pathways:

Pathway	Description	Example
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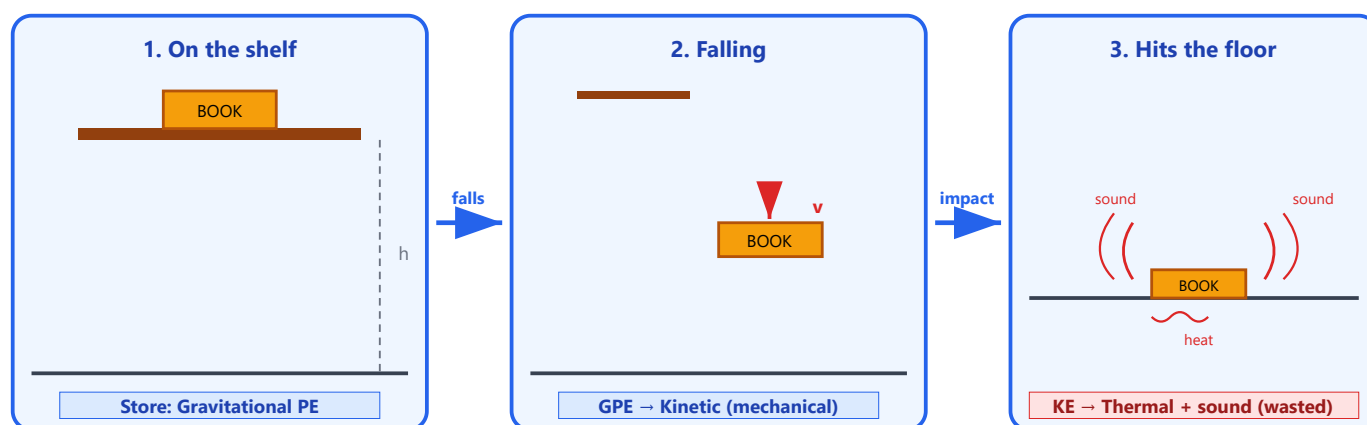
Mechanically	A force does work on an object over a distance.	Pushing a box along the floor (chemical → kinetic + thermal).
Electrically	A current flows through a circuit — charges do work.	A kettle heating water (chemical at power station → thermal of water).
By heating	Energy flows from a hotter to a colder region by conduction, convection, or radiation.	Hot coffee cooling down in a cold room (thermal of coffee → thermal of surroundings).
By radiation	Energy carried by waves — light, sound, infrared, other EM waves.	Sunlight warming the ground (nuclear in Sun → thermal of ground).

Principle of Conservation of Energy

Energy cannot be created or destroyed. It can only be transferred usefully, stored, or dissipated. In a **closed system**, the total energy before any event is equal to the total energy after the event — the energy has simply moved between stores.

A **closed system** is one in which no energy (and no matter) enters or leaves the boundary of the system. In practice, truly closed systems are rare — almost every real process loses some energy to the surroundings as thermal energy — but we treat systems as closed when the boundary is carefully chosen (e.g., a sealed, insulated flask).

Energy stores and transfers — a falling book



Throughout the whole event, the **total energy stays the same**. At the start, all the energy is in the gravitational potential store. In mid-fall, some is GPE and some is kinetic. At the moment of impact, virtually all of it has been transferred to the thermal stores of the floor, the book, and the surrounding air, plus a little carried away as a sound wave. Add it all up — you get back the original GPE.

Key Equations in 4.1.1

You must know and be able to apply all of these. AQA provides only $E_e = \frac{1}{2} k e^2$ and $\Delta E = m c \Delta \theta$ on the equation sheet — E_k , ΔE_p and $W = F d$ must be recalled.

Kinetic energy: $E_k = \frac{1}{2} m v^2$

where E_k = kinetic energy in joules (J), m = mass in kilograms (kg), v = speed in metres per second (m/s).

Gravitational potential energy: $\Delta E_p = m g h$

where ΔE_p = change in GPE in joules (J), m = mass in kg, g = gravitational field strength (9.8 N/kg on Earth), h = change in height in metres (m).

Elastic potential energy: $E_e = \frac{1}{2} k e^2$

where E_e = elastic potential energy in joules (J), k = spring constant in newtons per metre (N/m), e = extension in metres (m). Assumes the spring has not been stretched beyond its limit of proportionality.

Work done: $W = F d$

where W = work done in joules (J), F = force in newtons (N), d = distance moved in the direction of the force in metres (m). 1 joule = 1 newton-metre.

Thermal energy (specific heat capacity): $\Delta E = m c \Delta \theta$

where ΔE = energy transferred in joules (J), m = mass in kg, c = specific heat capacity in

J/kg °C, $\Delta\theta$ = temperature change in °C (or K — it is the same size of interval).

Worked Example 1 — Kinetic Energy

Question: A 1500 kg car is driving at 15 m/s. Calculate its kinetic energy.

Given: $m = 1500 \text{ kg}$, $v = 15 \text{ m/s}$

Formula: $E_k = \frac{1}{2} m v^2$

Working: $E_k = 0.5 \times 1500 \times 15^2$

$$E_k = 0.5 \times 1500 \times 225$$

$$E_k = 750 \times 225$$

$$E_k = 168\,750 \text{ J} \quad (\text{or } 168.75 \text{ kJ})$$

Answer: 168 750 J (to 5 s.f.)

Notice that because speed is **squared**, doubling the speed gives *four* times the kinetic energy. This is why braking distances grow so rapidly at high speed.

Worked Example 2 — Gravitational Potential Energy

Question: A 70 kg person climbs a 2 m ladder. How much gravitational potential energy do they gain?

Given: $m = 70 \text{ kg}$, $g = 9.8 \text{ N/kg}$, $h = 2 \text{ m}$

Formula: $\Delta E_p = m g h$

Working: $\Delta E_p = 70 \times 9.8 \times 2$

$$\Delta E_p = 1372 \text{ J}$$

Answer: 1372 J

Worked Example 3 — Energy Conservation (Falling Object)

Question: A 0.5 kg ball is dropped from a height of 1.8 m.

Assuming no air resistance, what is its speed just before it hits the ground?

Strategy: All GPE is transferred to KE (closed system, no losses).

$$\Delta E_p = E_k \rightarrow mgh = \frac{1}{2}mv^2$$

$$\begin{aligned}v^2 &= 2gh \\v &= \sqrt{2 \times 9.8 \times 1.8} \\v &= \sqrt{35.28} \\v &= 5.94 \text{ m/s}\end{aligned}$$

Answer: 5.94 m/s (to 3 s.f.)

Exam Tip: Always check your **units**. Mass must be in kg (not g), speed in m/s (not km/h or mph), and height in metres (not cm). A common error is forgetting to square v in the kinetic energy formula. Another is using $g = 10 \text{ N/kg}$ when AQA specifically states **9.8 N/kg** for GCSE Physics (separate science). Show your working line-by-line — method marks are awarded even if the final answer is wrong.

4.1.2 Conservation and Dissipation of Energy

In reality, no energy transfer is 100% efficient — some energy is always **dissipated** (spread out) to the surroundings, usually as **thermal energy**. This wasted energy is still there — it has not been destroyed — but it is no longer useful for the purpose we wanted.

Useful energy: the energy transferred to the store(s) we want.

Wasted (dissipated) energy: energy transferred to stores we did not want — usually the thermal stores of the surroundings, and often some sound.

Examples of useful vs wasted energy transfers:

- **Electric kettle:** useful = thermal store of water. Wasted = thermal store of the kettle body, sound of bubbling.
- **Car engine:** useful = kinetic store of the car. Wasted = thermal (hot engine, exhaust gases) and sound.
- **Filament light bulb:** useful = light (radiation). Wasted = thermal — around 95% of the input electrical energy is dissipated as heat, which is why filament bulbs have largely been replaced by LEDs.
- **Loudspeaker:** useful = sound (radiation). Wasted = thermal from the coil and magnet.

Efficiency

Efficiency is the proportion of input energy that is transferred usefully.

$$\text{efficiency} = \text{useful output energy transfer} \div \text{total input energy transfer}$$

It can also be calculated using power:

$$\text{efficiency} = \text{useful output power} \div \text{total input power}$$

Efficiency is a ratio, so it has no units. It is expressed as a decimal (0 to 1) or multiplied by 100 to give a percentage (0% to 100%).

Worked Example 4 — Efficiency

Question: An electric motor is supplied with 100 J of electrical energy. It transfers 60 J usefully as kinetic energy. Calculate its efficiency.

Given: Useful output = 60 J, Total input = 100 J

Formula: $\text{efficiency} = \text{useful output} \div \text{total input}$

Working: $\text{efficiency} = 60 \div 100$
 $= 0.6$
 $= 60\%$

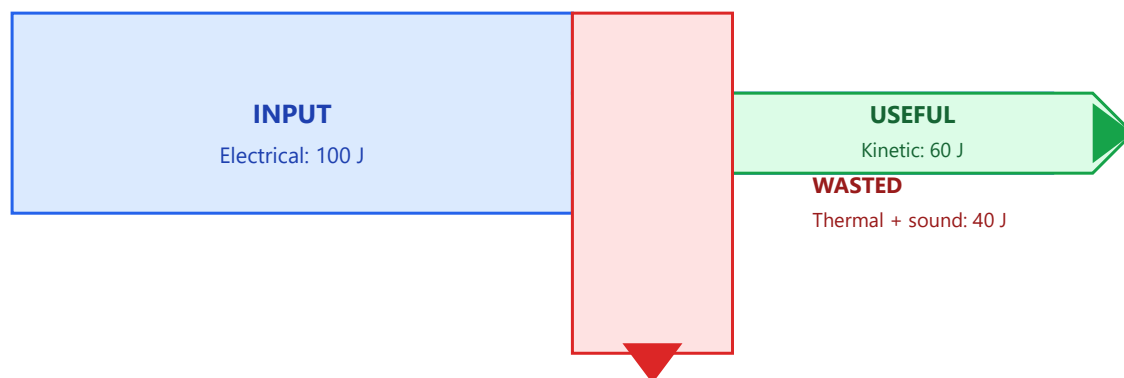
Where did the other 40 J go?

→ 40 J dissipated as thermal energy (friction in bearings, heating of coils) and some as sound.

Sankey Diagrams

A **Sankey diagram** shows energy transfers visually. The width of each arrow is proportional to the amount of energy it represents. Wasted branches peel off to the side (usually downwards) — the remaining useful energy continues to the right. Sankey diagrams make it easy to see at a glance how efficient a device is.

Sankey diagram — 60% efficient motor



Arrow widths are proportional to energy — 40 J wasted is 40% of the input width, 60 J useful is 60%.

Reducing Unwanted Energy Transfers

Because dissipated energy is wasted, engineers go to great lengths to reduce it. You must know at least the following methods:

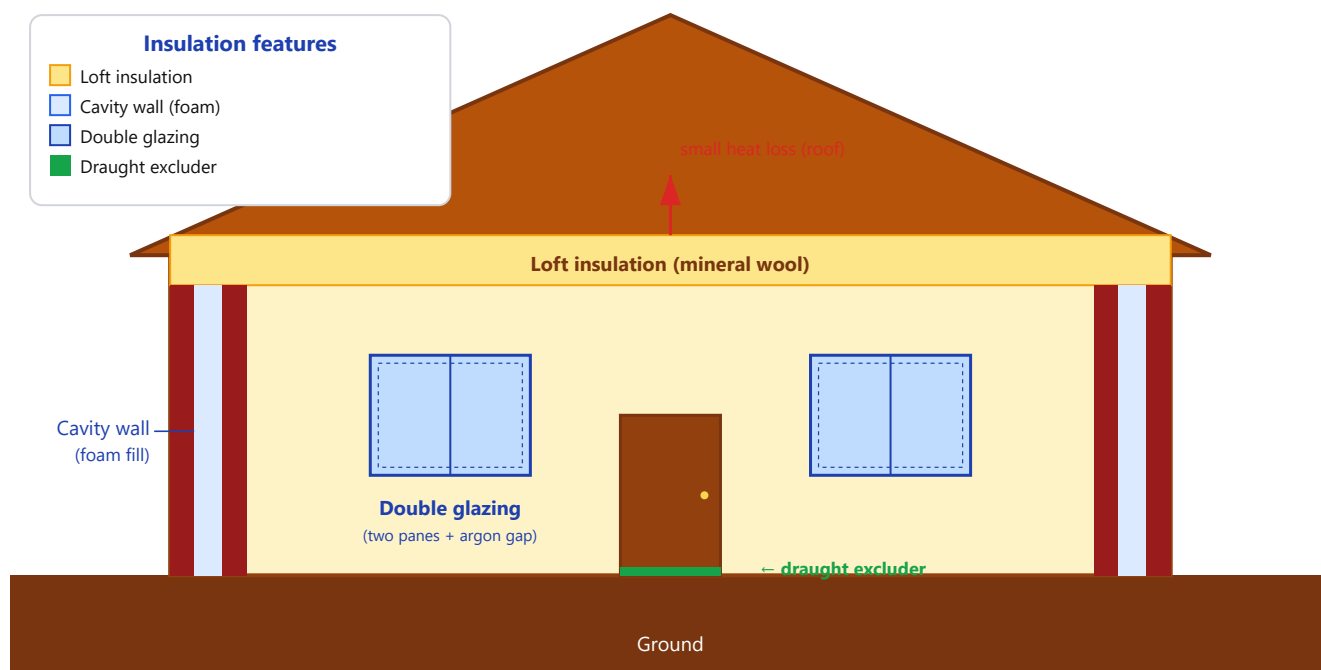
Problem	Solution	How it works
Friction between moving parts	Lubrication (e.g., engine oil, grease, graphite)	A thin film of oil between surfaces reduces the force of friction, so less work is done against friction and less energy is dissipated as thermal energy. Also reduces wear on the parts.
Air resistance / drag on vehicles	Streamlined / aerodynamic shape	Smooth, curved shapes let air flow around the object with less turbulence. Less force needed to push through the air — less energy wasted as thermal energy of the air.
Heat loss from a house through walls	Cavity wall insulation — filling the gap between two brick walls with foam or mineral wool	The trapped air in the insulation has a very low thermal conductivity. Heat escapes much more slowly by conduction.
Heat loss through the roof	Loft insulation (glass wool / mineral wool laid between	Traps pockets of air (low thermal conductivity) above the ceiling. Reduces

	joists)	conduction and convection losses — hot air rises, so a lot of heat (about a quarter) is lost through the roof of a poorly insulated house.
Heat loss through windows	Double glazing — two panes of glass with a gap filled with a low-conductivity gas (argon) or vacuum	The gas has a much lower thermal conductivity than glass alone — heat conducts through far more slowly. A vacuum is even better because there are no particles to conduct at all.
Heat loss through gaps around doors/windows	Draught excluders	Seal the gaps so warm air cannot leave by convection and cold air cannot enter.

Thermal conductivity is a measure of how quickly a material conducts thermal energy. A material with **low** thermal conductivity (e.g., mineral wool, foam, air) is a good **insulator** — it transfers heat slowly. A material with **high** thermal conductivity (e.g., copper, aluminium) is a good **conductor**.

The rate of cooling of a building is affected by the **thickness** and **thermal conductivity** of its walls. Thicker walls with lower-conductivity materials lose heat more slowly — the building stays warm for longer once heated.

Thermal insulation in a house — cross-section



Exam Tip: When asked about reducing heat loss, state *both* the method **and** the physics. For example: "Cavity wall insulation fills the gap between the two walls with foam. The foam has a low thermal conductivity, so energy is transferred by conduction more slowly, reducing heat loss." Just saying "cavity wall insulation" on its own will score only 1 mark instead of 2.

4.1.3 Specific Heat Capacity

Different substances need different amounts of energy to warm them up by the same amount. Water, for instance, needs a great deal of energy to heat — this is why oceans moderate climate so effectively.

The **specific heat capacity** of a substance is the amount of energy required to raise the temperature of **1 kg** of the substance by **1 °C**.

Symbol: c . Units: $\text{J/kg } ^\circ\text{C}$ (joules per kilogram per degree Celsius).

Equation: $\Delta E = m c \Delta \theta$

where ΔE = change in thermal energy (J), m = mass (kg), c = specific heat capacity (J/kg °C), $\Delta \theta$ = change in temperature (°C).

The Greek letter θ (theta) is used to represent temperature in AQA Physics, and Δ (delta) means "change in". So $\Delta \theta$ means "change in temperature" — final temperature minus initial temperature.

Typical Specific Heat Capacities

Substance	c (J/kg °C)	Notes
Water	4200	Unusually high — why oceans take a long time to warm up and cool down.
Ice	2100	Solid water has half the heat capacity of liquid water.
Oil (cooking)	2000	Heats up faster than water — useful in frying.
Concrete	880	Used in night-storage heaters.
Aluminium	900	Pans made of aluminium heat up quickly.
Iron	450	Heats up about 9× faster than the same mass of water.
Copper	385	Common material for heating pipes — heats quickly, conducts well.
Lead	130	Very low heat capacity — warms very quickly for its mass.

Worked Example 5 — Heating Water

Question: How much energy is needed to heat 2 kg of water from 20 °C to 80 °C? ($c_{\text{water}} = 4200 \text{ J/kg } ^\circ\text{C}$)

Given: $m = 2 \text{ kg}$, $c = 4200 \text{ J/kg } ^\circ\text{C}$, $\Delta\theta = 80 - 20 = 60 \text{ } ^\circ\text{C}$

Formula: $\Delta E = m c \Delta\theta$

Working: $\Delta E = 2 \times 4200 \times 60$
 $= 504\,000 \text{ J}$
 $= 504 \text{ kJ}$

Answer: $504\,000 \text{ J}$ (504 kJ)

Worked Example 6 — Rearranging for Temperature Change

Question: A 0.5 kg block of aluminium is given 9000 J of thermal energy. What is its temperature rise? ($c = 900 \text{ J/kg } ^\circ\text{C}$)

Given: $m = 0.5 \text{ kg}$, $c = 900 \text{ J/kg } ^\circ\text{C}$, $\Delta E = 9000 \text{ J}$

Formula: $\Delta E = m c \Delta\theta \rightarrow \Delta\theta = \Delta E \div (m c)$

Working: $\Delta\theta = 9000 \div (0.5 \times 900)$
 $= 9000 \div 450$
 $= 20 \text{ } ^\circ\text{C}$

Answer: Temperature rises by $20 \text{ } ^\circ\text{C}$.

Why Water's High Specific Heat Capacity Matters

Water has a very high specific heat capacity ($4200 \text{ J/kg } ^\circ\text{C}$) compared with most other common substances. Practical consequences include:

- **Climate:** oceans absorb huge amounts of energy from the Sun without warming much. Coastal climates are therefore milder — summers are cooler and winters are warmer than inland.
- **Storage heaters:** night-storage heaters are filled with concrete or water to store lots of thermal energy overnight (using cheap off-peak electricity) and release it slowly during the day.
- **Central heating:** water is pumped through radiators because it carries a lot of energy per kilogram.
- **Cooling systems:** car engines use water-based coolant because it can absorb a great deal of thermal energy before boiling.
- **Living organisms:** because our bodies are mostly water, we resist rapid temperature changes — this is vital for homeostasis.

Exam Tip: A common exam mistake is confusing **specific heat capacity** (energy to warm 1 kg of a substance by 1 °C) with **specific latent heat** (energy to change the state of 1 kg without temperature change). They are *different*. For GCSE separate physics, **c** only applies while the substance stays in the same state — during melting or boiling, temperature is constant and you need the latent-heat equation instead.

4.1.4 Power

Power is the rate at which energy is transferred, or equivalently the rate at which work is done.

$$P = E / t \quad \text{or} \quad P = W / t$$

where **P** = power in **watts (W)**, **E** or **W** = energy transferred / work done in joules (J), **t** = time in seconds (s).

The **watt** is named after James Watt, the 18th-century Scottish engineer who improved the steam engine. **1 watt = 1 joule per second**. A 100 W light bulb transfers 100 joules of energy every second.

Useful multiples:

- **1 kilowatt (kW) = 1000 W** — a typical electric kettle is around 2–3 kW.
- **1 megawatt (MW) = 1 000 000 W** — a single wind turbine is typically 2–5 MW.
- **1 gigawatt (GW) = 10⁹ W** — large nuclear power stations output 1–3 GW.

Worked Example 7 — Energy Transferred by a Light Bulb

Question: A 60 W light bulb is left on for 5 minutes. How much energy is transferred?

Given: $P = 60 \text{ W}$, $t = 5 \text{ min} = 300 \text{ s}$

Formula: $P = E / t \rightarrow E = P \times t$

Working: $E = 60 \times 300$
 $= 18\,000 \text{ J}$
 $= 18 \text{ kJ}$

Answer: 18 000 J (18 kJ)

Worked Example 8 — Power of a Person Climbing Stairs

Question: A person of mass 60 kg climbs 30 steps, each 0.2 m high, in 10 seconds. Calculate (a) the work done against gravity, and (b) their average power.

Given: $m = 60 \text{ kg}$, number of steps $\times$ height $= 30 \times 0.2 = 6 \text{ m}$
 $g = 9.8 \text{ N/kg}$, $t = 10 \text{ s}$

(a) Work done = force $\times$ distance = weight $\times$ height gained

$$\text{Weight} = m \times g = 60 \times 9.8 = 588 \text{ N}$$

$$W = F \times d = 588 \times 6 = 3528 \text{ J}$$

(b) Power = energy / time

$$P = 3528 \div 10$$

$$= 352.8 \text{ W}$$

Answer: (a) 3528 J (b) 352.8 W ($\approx 350 \text{ W}$)

For comparison: a professional cyclist sustains $\sim 400 \text{ W}$ for an hour; an elite sprinter peaks briefly above 2000 W .

Exam Tip: If a question asks for the **useful** power output, make sure you use the useful energy output — not the total energy input. If the device is 60% efficient and uses 100 W from the mains, its useful power output is only 60 W . Units: always convert minutes and hours to seconds before substituting into $P = E / t$. Forgetting this is a very common source of lost marks.

4.1.5 National and Global Energy Resources

Energy resources are used to generate electricity, for transport, and for heating. They fall into two broad categories:

Renewable resources are replenished at the same rate (or faster) than they are used. They will not run out on any human timescale.

Non-renewable resources have finite supplies that will eventually be exhausted because they are used far faster than they form.

Non-Renewable Resources

Resource	Main uses	How it works	Advantages	Disadvantages
Coal	Electricity, (historically) heating	Burned in a boiler to heat water into steam, which turns turbines coupled to generators.	High energy density; available in many countries; cheap; reliable — can be burned on demand.	Releases large quantities of CO ₂ (greenhouse gas) and SO ₂ (causes acid rain); mining damages landscape; finite resource.
Oil	Transport (petrol, diesel), heating, plastics	Refined into fuels burned in engines and boilers; also a feedstock for plastics and chemicals.	Extremely high energy density; easy to transport (liquid); the backbone of most modern transport.	Releases CO ₂ ; oil spills damage ecosystems; finite; geopolitical supply risks.
Natural gas	Electricity, domestic heating, cooking	Burned in a boiler (like coal) or in a combined-cycle gas turbine (CCGT), which is more efficient.	Cleaner than coal (~half the CO ₂ per kWh); flexible — can ramp up and down quickly to match demand; widely available via pipelines.	Still releases CO ₂ ; methane leaks are a very potent greenhouse gas; finite.

Nuclear	Electricity	Fission of uranium-235 releases energy as thermal energy; the heat boils water and drives steam turbines.	No CO ₂ during generation; very high energy density (1 kg of U-235 $\approx$ 2.5 million kg of coal); reliable baseload.	Radioactive waste must be stored safely for thousands of years; high decommissioning costs; risk of accidents (Chernobyl, Fukushima); very high build cost.
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Renewable Resources

Resource	Main uses	How it works	Advantages	Disadvantages
Wind	Electricity	Wind turns the blades of a turbine, driving a generator.	No CO ₂ during generation; low running costs; plentiful in UK.	Unreliable — no output when the wind is not blowing; visual/noise impact; hazard to birds; high capital cost.
Solar	Electricity (photovoltaic), hot water (solar thermal)	PV panels convert sunlight directly into electricity via the photoelectric effect in semiconductor cells.	No CO ₂ in use; very low maintenance; ideal for remote locations; abundant.	No output at night or when heavily overcast; low power density (big area needed); manufacture requires rare materials.
Hydroelectric	Electricity	Water stored behind a dam falls through turbines (GPE $\rightarrow$ KE $\rightarrow$ electrical).	Reliable in rainy regions; can quickly respond to demand; pumped storage	Flooding valleys destroys habitats and displaces people; only viable in certain

			can store surplus energy.	geography; dam failure risk.
Tidal	Electricity	Water is trapped behind a tidal barrage at high tide; released through turbines as the tide falls.	Tides are extremely predictable — perfectly reliable; no fuel cost; long-lived infrastructure.	Very limited number of suitable sites; disrupts estuarine ecosystems; high build cost; generates only at tide changes.
Wave	Electricity	The up-and-down motion of ocean waves drives mechanisms that turn generators.	No CO ₂ ; UK has extensive coastline; waves are a concentrated form of solar/wind energy.	Output depends on sea state; harsh marine environment erodes equipment; still an emerging technology; visual impact.
Geothermal	Electricity, domestic heating	Heat from radioactive decay deep within the Earth boils water into steam to drive turbines.	Very reliable — runs 24/7 regardless of weather; small land footprint; low CO ₂ .	Only viable where hot rocks are close to the surface (Iceland, NZ, parts of US); can release small amounts of gas from deep rocks; drilling is expensive.
Biofuel	Transport, heating, electricity	Crops (e.g., sugar cane, corn, wood) are burned or fermented. CO ₂ released was recently absorbed	"Carbon neutral" in principle; fuel can be stored and burned on demand; supports	Competes with food crops for land; deforestation to grow biofuels; intensive farming requires fertiliser;

		from the atmosphere.	farming economies.	still emits some CO ₂ when burned.
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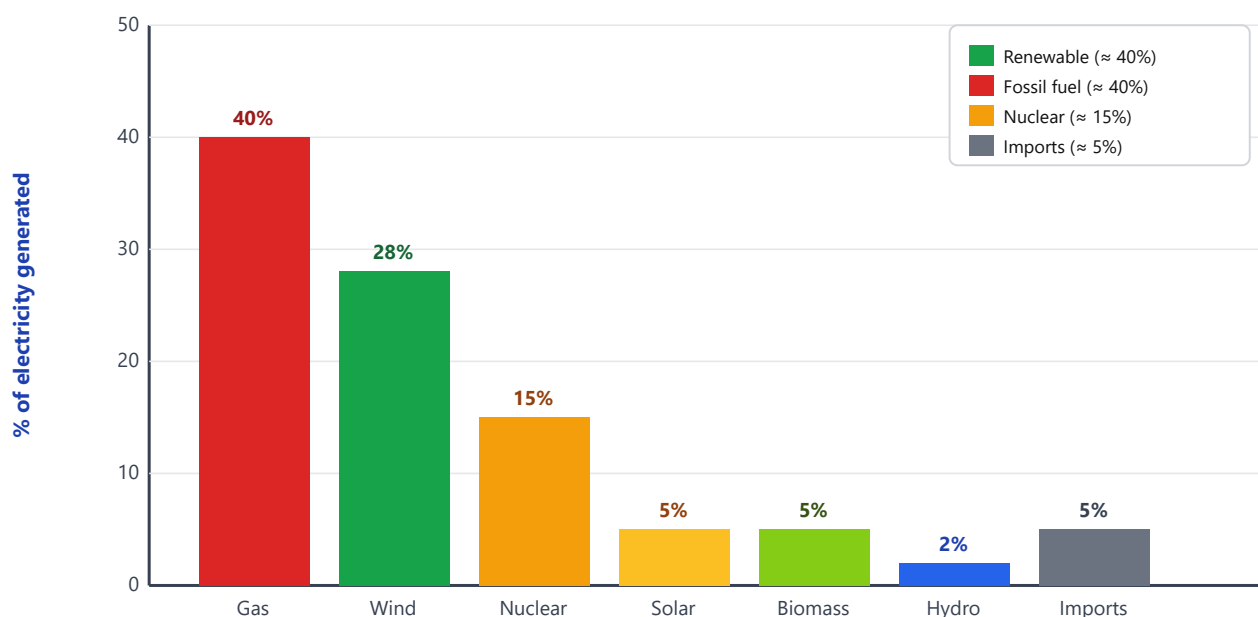
Summary Comparison

Resource	Type	Reliability	CO ₂ ?	Cost	Typical uses
Coal	Non-renewable	High	Very high	Low	Electricity
Oil	Non-renewable	High	High	Medium	Transport
Gas	Non-renewable	High	Medium	Medium	Electricity, heating
Nuclear	Non-renewable	Very high (baseload)	None in use	High build, low fuel	Electricity
Wind	Renewable	Low (intermittent)	None	Medium	Electricity
Solar	Renewable	Low (daytime only)	None	Medium/falling	Electricity, hot water
Hydro	Renewable	High	None	High build	Electricity
Tidal	Renewable	Very high (predictable)	None	Very high	Electricity
Wave	Renewable	Medium	None	High	Electricity
Geothermal	Renewable	Very high	Low	High	Electricity, heating
Biofuel	Renewable	High	"Neutral"	Medium	Transport, heating

UK Electricity Generation Mix — Changing Over Time

The UK's electricity mix has changed dramatically since 1990. Coal used to supply about 65% of the UK's electricity; today it has been phased out altogether. Gas and renewables (especially wind) have replaced it. The UK generated its last coal-fired electricity in September 2024 — the first G7 country to phase out coal entirely.

Approximate UK electricity generation mix (recent)



Figures are approximate — proportions change year to year. Coal has been phased out (2024).

Environmental Issues

Different energy sources cause different environmental problems:

- **Burning fossil fuels (coal, oil, gas)** releases **carbon dioxide (CO₂)**, a greenhouse gas that traps thermal energy radiated from Earth's surface, causing **global warming** and **climate change**.
- **Burning coal** (and to a lesser extent oil) releases **sulfur dioxide (SO₂)**, which dissolves in rain to form **acid rain**, damaging forests, lakes, and stonework.
- **Oil spills** from tankers or rigs devastate marine and coastal wildlife and take decades to clean up.
- **Nuclear power** produces **radioactive waste** that remains dangerous for thousands of years and must be stored safely. Accidents (Chernobyl 1986, Fukushima 2011) can contaminate large areas.
- **Hydroelectric dams** flood valleys, destroying habitats and displacing communities; they can also release methane from decaying vegetation.

- **Wind farms** have visual and noise impacts and can kill birds and bats, though the toll is small compared with cats or buildings.
- **Biofuels** compete with food crops and may drive deforestation if rainforests are cleared for plantations.
- **Solar panel manufacture** uses energy and rare materials; end-of-life recycling is still developing.

Reliability vs Environmental Impact — the Energy Trilemma

Policymakers must balance three competing demands, often called the **energy trilemma**:

1. **Security & reliability** — the lights must stay on even when there is no wind and no sun.
2. **Affordability** — electricity must not be so expensive that consumers or industry suffer.
3. **Sustainability** — reducing CO₂ emissions to meet climate targets.

No single resource satisfies all three perfectly. A realistic low-carbon electricity system uses a **mix** of resources: renewables for most generation, nuclear for reliable low-carbon baseload, gas for flexible backup during calm/dark periods, and increasingly large-scale storage (batteries, pumped hydro) to shift energy from times of surplus to times of shortage.

Exam Tip: Long-answer ("6-mark") questions about energy resources almost always reward a **balanced** answer — you must discuss both advantages and disadvantages, consider environmental *and* economic impact, and reach a justified conclusion. A one-sided answer ("renewables are better, end of story") will cap at a middling grade. Always write: "advantage... however, disadvantage... overall..." with specific physics reasoning.

Exam Tip: Know the difference between **energy resource** (the raw source, e.g. coal, uranium, wind) and the **generation method** (how electricity is produced, e.g. boiler + steam turbine, PV panel). Also know that *all* generation methods except PV and fuel cells involve turning a turbine connected to a generator — the only real difference between a coal, nuclear, gas, wind, tidal, and geothermal power station is *what* turns the turbine.

Summary Checklist for 4.1 Energy

- Name the 8 energy **stores** and give an example of each.
- Name the 4 energy **pathways** (transfers) and give an example of each.
- State and apply the principle of conservation of energy in a closed system.
- Calculate kinetic energy using $E_k = \frac{1}{2} m v^2$.
- Calculate gravitational PE using $\Delta E_p = m g h$ with $g = 9.8 \text{ N/kg}$.
- Calculate elastic PE using $E_e = \frac{1}{2} k e^2$.
- Calculate work done using $W = F d$.
- Calculate thermal energy changes using $\Delta E = m c \Delta \theta$.
- Define useful and wasted energy, calculate efficiency, and read/draw a Sankey diagram.
- Describe ways of reducing unwanted energy transfer: lubrication, streamlining, loft insulation, cavity walls, double glazing, draught excluders — and explain the physics of each.
- Define specific heat capacity and use the data in calculations.
- Define power and calculate it using $P = E / t$.
- Compare renewable and non-renewable resources, their advantages, disadvantages, and environmental impacts.
- Discuss the changing UK energy mix and the energy trilemma in 6-mark extended-answer questions.